Supporting Information

Random cross-validation understates cross-cohort calibration error in multi-cohort proteomics: an accession-aligned evaluation-protocol contrast

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S1. Scope and corrected identities

This supplement accompanies the accession-aligned six-cohort analysis. Its tables derive from the same stored predictions used by the article figures. Historical submission versions and historical gene-indexed analyses are not updated copies of this supplement.

The 1,191 samples comprise 689 tumors, 496 adjacent-normal samples and six HNSCC cored-normal samples. The latter are identified by the source suffix `.C` and, like `.N` samples, have label zero. Both suffixes map to the corresponding unsuffixed patient identifier, with cohort included in the patient key. The resulting 719 patients include 470 with both tissue classes, 219 tumor-only and 30 normal-only patients. HNSCC contributes 110 tumors and 72 normals from 112 patients. Normal here means a source-defined nonmalignant sample, not a healthy donor. Patients with tumor, adjacent-normal and cored-normal samples stay together under the grouped split.

The sample manifest records cohort, sample ID, tissue class, binary label, true patient ID, and fold for each of the three protocols. Sample CV splits 381 patients across evaluation folds. Patient-grouped CV and LOGO split zero true patients. The corrected GroupKFold partition was generated in scikit-learn 1.9.0 and is preserved by sample identity; it is not the historical partition based on 723 parsed keys.

The primary protein intersection comprises 8,906 stable ENSP accessions. Version suffixes are stripped only after verifying unique stable accessions per cohort; gene annotation and, for the retained set, the complete accession version also agree across all cohorts. There is no gene-level averaging in this primary matrix. An old 9,675-gene intersection selected the first protein for each name; 1,234 names selected different stable accessions in different cohorts. The 8,441-name sensitivity retains only the consistent first-accession subset. Neither mapping guarantees absence of every protein-quantification ambiguity, but the primary matrix no longer equates different stable accessions merely because they share a gene name.

S2. Aggregation and probability-error definitions

AUROC, ten-bin ECE, Brier score and log-loss are computed within each source cohort on its stored out-of-fold predictions. Equal-cohort macro values average those six metrics. Pooled values concatenate all predictions before computing the metric. For sample and patient CV, within-cohort metrics combine predictions from different CV models; for LOGO, each cohort is scored by one model trained on the other five. The pooled AUROC additionally includes cross-cohort positive–negative pairs. It is not the mean of within-cohort or CV-fold AUROCs.

The Brier score is $n^{-1}\sum_i (p_i - y_i)^2$. Log-loss is $-n^{-1}\sum_i [y_i \log p_i + (1 - y_i)\log(1 - p_i)]$, with numerical clipping provided by the recorded metric implementation. Ten-bin ECE weights the absolute mean-probability/tumor-fraction discrepancy by each bin's sample mass. No causal meaning is assigned to an AUROC increment or to the AUROC–ECE association across these six confounded cohorts.

S3. Cohort-resolved primary and comparator results

The tables below contain one held-out cohort per LOGO fold. ECE uses ten equal-width probability bins; Brier and log-loss are unbinned complementary summaries. No historical confidence intervals are attached to these corrected predictions.

Table S1: Primary accession-aligned XGBoost abundance predictions.

Cohort	AUROC	ECE	Brier	Log-loss
CCRCC	0.8485	0.2522	0.2245	0.8162
LUAD	0.9998	0.0203	0.0064	0.0253
UCEC	0.9184	0.1501	0.1376	0.5270
HNSCC	0.9809	0.0368	0.0438	0.1656
LSCC	0.9974	0.0057	0.0048	0.0338
PDAC	0.8610	0.2506	0.2533	0.7602

Table S2: Three reference-normalized global abundance summaries: observed mean, median and detection fraction. These are not raw total intensity or an identified biological component.

Cohort	AUROC	ECE	Brier	Log-loss
CCRCC	0.9237	0.0507	0.1126	0.3556
LUAD	0.9618	0.0509	0.0685	0.2440
UCEC	0.6410	0.1433	0.2579	0.7879
HNSCC	0.7824	0.0909	0.1851	0.5636
LSCC	0.9936	0.0990	0.0457	0.1783
PDAC	0.6004	0.2201	0.2869	0.8707

Table S3: Missingness-only XGBoost predictions on the same accession matrix. Weak within-cohort ranking and pooled ranking are distinct.

Cohort	AUROC	ECE	Brier	Log-loss
CCRCC	0.5721	0.2400	0.2985	0.8117
LUAD	0.5243	0.0525	0.2529	0.6994
UCEC	0.5790	0.1749	0.2458	0.6853
HNSCC	0.5585	0.1738	0.2694	0.7342
LSCC	0.5430	0.1517	0.2738	0.7477
PDAC	0.4900	0.1614	0.2762	0.7491

S4. Mapping, model-family and imputation sensitivities

These analyses use the corrected tissue and patient identities. The mapping sensitivity changes the feature set; the L2 and deliberately leaky-median arms retain the primary 8,906 accessions and six LOGO folds. They are sensitivities on the same source cohorts, not independent replications.

Table S4: The 8,441 first-accession-consistent-gene mapping sensitivity. No inconsistent historical gene/protein match is retained.

Cohort	AUROC	ECE	Brier	Log-loss
CCRCC	0.8166	0.2757	0.2548	0.9372
LUAD	0.9996	0.0167	0.0069	0.0267
UCEC	0.9380	0.1526	0.1362	0.5168
HNSCC	0.9814	0.0242	0.0339	0.1485
LSCC	0.9972	0.0051	0.0048	0.0343
PDAC	0.8647	0.2384	0.2267	0.6594

Table S5: L2 logistic regression with train-fold median imputation and train-fold standardization.

Cohort	AUROC	ECE	Brier	Log-loss
CCRCC	0.9037	0.1164	0.1327	0.7712
LUAD	1.0000	0.0132	0.0033	0.0155
UCEC	0.9624	0.0794	0.0746	0.2805
HNSCC	0.9866	0.0503	0.0507	0.2585
LSCC	0.9952	0.0038	0.0048	0.0563
PDAC	0.7567	0.2409	0.2539	0.9380

Table S6: XGBoost with deliberately full-dataset medians, including held-out feature values but no held-out labels. This is a diagnostic, not the recommended evaluation.

Cohort	AUROC	ECE	Brier	Log-loss
CCRCC	0.8510	0.2740	0.2443	0.8676
LUAD	0.9997	0.0191	0.0073	0.0283
UCEC	0.9114	0.1540	0.1433	0.5483
HNSCC	0.9806	0.0261	0.0378	0.1512
LSCC	0.9954	0.0058	0.0050	0.0378
PDAC	0.8577	0.2539	0.2551	0.7743

S5. Conditional paired-patient resampling

True patients are sampled with replacement within each of the six fixed cohorts, carrying all of each patient's sample predictions together. The same patient draw is used for sample CV, patient CV and LOGO, so each contrast is paired. We use 2,000 draws and NumPy random seed 20260922. The stored model predictions are not refitted.

The 2.5th–97.5th percentile ranges in Table S7 describe sensitivity to the observed patient mixture conditional on those predictions and cohorts. They exclude retraining uncertainty, model-selection variability and variation in future study environments. The source cohorts are not asserted to be exchangeable random studies. Thus these ranges are not full algorithm-refit or future-study confidence intervals.

Table S7: Paired protocol AUROC contrasts and conditional percentile ranges.

Aggregation	Contrast	Point	2.5th–97.5th percentiles
Macro	sample minus logo	0.0565	0.0442–0.0703
Macro	patient minus logo	0.0572	0.0448–0.0708
Macro	sample minus patient	-0.0007	-0.0032–0.0016
Pooled	sample minus logo	0.0386	0.0305–0.0478
Pooled	patient minus logo	0.0389	0.0306–0.0482
Pooled	sample minus patient	-0.0003	-0.0015–0.0010

S6. Linear model and recalibration implementation

The L2 classifier is `LogisticRegression` with $C = 1$, `solver=lbfgs`, L2 penalty and `max_iter=500`. Missing values use training-fold medians and scaling parameters are estimated on the training fold only. All six fits finished without a convergence warning (Table S8). The superseded method-lock text giving 1000 iterations does not describe these fits.

Table S8: Recorded convergence of the six L2 LOGO fits.

Held-out cohort	Iterations	Limit	Converged
CCRCC	15	500	Yes
HNSCC	15	500	Yes
LSCC	44	500	Yes
LUAD	43	500	Yes
PDAC	12	500	Yes
UCEC	15	500	Yes

For recalibration, the recorded cohort order is CCRCC, LUAD, UCEC, HNSCC, LSCC and PDAC. After removing the outer test cohort, the first two remaining cohorts and the other three form the inner groups. Training on one and scoring the other, then swapping, yields the calibration data. Inner training-fold imputation is separate from the outer fit. Temperature scaling uses probability logits clipped to $[10^{-7}, 1 - 10^{-7}]$ before transformation, and bounded NLL minimization over $T \in [0.05, 20]$. Platt scaling uses logistic regression on those logits ($C = 10^6$, maximum 2000 iterations). The maps never use outer held-out labels.

The inner models train on two or three cohorts, unlike the outer five-cohort predictor. This training-size mismatch limits generalization of any conclusion about recalibration. Distinct cohort-specific maps can also change cross-cohort pooled rankings while preserving each within-cohort monotone ranking. The main article reports the observed benefits and losses of this specific procedure without a universal claim.

S7. Side-study dependency disposition

The following distinctions prevent a historical computation from being mistaken for evidence on the corrected matrix. Archived files remain available; removal of their numerical claims from the current argument does not imply that their scientific questions should be abandoned.

Corrected matrix sensitivities retained. The 8,441-feature mapping comparison, full-data-median XGBoost, train-fold L2, coverage curves, and split-half recalibration were rerun on corrected labels and aligned proteins, and verified against their stored prediction rows. Their results appear in the article and this supplement.

OV plex-LOGO retained as historical summary context. The ovarian table and TMT annotation are independent of the six-cohort intersection. Their stored source checksums and the 103-sample, 84-tumor/19-normal, 12-plex annotation inventory were checked during integration. The recorded pooled abundance AUROC is 0.985 under plex-LOGO and 0.987 under random CV; plex-LOGO ECE is 0.046 and global-summary AUROC is 0.951. There are only one to three normals per plex. These are summary-level historical results, not a new prediction-level replay. They are not promoted as an independently verified counterexample and do not deconfound the primary cancer/study comparison. The current six-cohort evidence itself does not support a universal gap. Historical bootstrap endpoints are not reused as uncertainty for the corrected six-cohort result.

Cohort-identity and label-permutation diagnostics not promoted. The old six-class identity classifier, one-versus-rest XGBoost check, training-cohort identity estimates, and within-/across-cohort label permutations use the old feature/label definitions. Their numerical claims require a corrected-object rerun before reuse. They are not needed to establish the observed primary protocol contrast.

Location/scale and representation diagnostics not promoted. Per-cohort median/MAD, mean/SD and ComBat-style location/scale restores, rank, quantile-normal, SVD whitening and PC-removal analyses depend on the previous matrix. Some additionally use held-out feature moments and therefore evaluate a target-feature-aware procedure. None of their historical numbers is presented as a corrected primary estimate.

Residual and injected-shift constructions not promoted. Within-cohort, within-class mean removal uses outcome labels and is a diagnostic construction, not a deployable held-out prediction procedure. Synthetic label-null injection estimates a response to a specified imposed shift. It is not a biological decomposition, and its previous target magnitude depended on an obsolete headline AUROC. Both would need redefinition and rerunning to support new numerical claims.

Two-way variance claim withdrawn. The earlier biological/cohort/interaction allocation does not identify biological and acquisition fractions in a confounded design. Its pseudo-interaction calculation is not retained as an interpretable two-way variance decomposition; rerunning the same formula on new data would not cure that problem.

BCM/UMICH dual-platform numbers not promoted. The three-cohort comparison uses a distinct gene-level mapping and same-patient join. It is neither an independent-patient validation nor a crossed cancer-by-batch design. A separate accession/aggregation and sample-identity audit is required before quantitative reuse of its effect-size correlations and LOGO estimates. No corrected six-cohort claim relies on those archived numbers.

Other acquisition-metadata screens. BRCA plex labels, the unavailable COAD mapping, and GBM/other cohort eligibility screens are historical source-inventory observations, not new predictive validation. No numerical BRCA or COAD performance claim is made. No pathway-enrichment analysis was performed.

S8. Reproducibility boundary

The corrected fits use `cptac` 1.5.14, XGBoost 3.3.0, NumPy 2.5.0, pandas 3.0.3 and scikit-learn 1.9.0. The primary model keeps the historical hyperparameter settings without an explicitly set XGBoost random state; sample stratification uses seed 28. Current figures and TeX tables derive from the verified stored predictions, so rendering

does not invoke a model fit. Hash checks establish version identity, not scientific validity.

The old submission directories are historical, unsynchronized packages. They must not be uploaded as this revised article. Full scientific clearance, any reinstatement of dependent side-study claims, and submission-package regeneration are separate from a successful local PDF build.